Lesson 1: Common Factors: The GCF of Monomials and Polynomials

Factoring is expanding run backwards — find the greatest common factor and remove it. (Extension: fractional common factors.)

1. Expanding and Factoring Are Opposite Processes

Multiplying polynomials out is called **expanding**: it turns a product into a sum or difference of terms.

monomial × polynomial:	$3x(x+4) = 3x^2 + 12x$
binomial × binomial → trinomial:	$(x+2)(x+6) = x^2 + 8x + 12$
	$(3x+2)(x+5) = 3x^2 + 17x + 10$
binomial × binomial → binomial:	$(x-7)(x+7) = x^2 - 49$

Factoring is the reverse: rewriting a sum or difference of terms as a product of factors. This unit builds a factoring toolbox one lesson at a time: **1** the greatest common factor; **2** binomial common factors and grouping; **3** $x^2 + bx + c$; **4** the difference of squares; **5** $ax^2 + bx + c$; **6** perfect squares and other special forms; **7** choosing a strategy; **8** unit review.

2. The Greatest Common Factor (GCF)

The greatest common factor of two whole numbers comes from their prime factorizations. For 60 and 84:

$$60 = 2 \times 2 \times 3 \times 5, \qquad 84 = 2 \times 2 \times 3 \times 7,$$

then multiply the prime factors common to both lists: GCF $= 2 \times 2 \times 3 = 12$. The same idea applies to monomials — for $12a^4$ and $18a^2b$:

$$12a^4 = 2 \times 2 \times 3 \times a \times a \times a \times a, \qquad 18a^2b = 2 \times 3 \times 3 \times a \times a \times b,$$

$$\text{GCF of } 12a^4 \text{ and } 18a^2b = 2 \times 3 \times a \times a = 6a^2.$$

Shortcut. By inspection: take the GCF of the numerical coefficients, then multiply by each common variable raised to its *lowest* exponent. So the GCF of $14p^2q^3$ and $21pq^4$ is $7pq^3$. A variable missing from even one monomial is not common, so it does not appear in the GCF. For a *set of monomials*, the GCF is positive unless *every* monomial is negative; then take the GCF negative too.

Example 1 — The GCF of a set of monomials

(a) Write the prime factorization of $18x^3y$ and $24xy^3$, and use it to determine the greatest common factor of $18x^3y$ and $24xy^3$.

(b) State the greatest common factor of each set of monomials by inspection.

(i) $16ab,\ 20a^2b^3$ **(ii)** $27x^4y^2,\ -36x^3y^5$ **(iii)** $a^2bc^3,\ 3abc$ **(iv)** $-30a^3b,\ -18a^2b^2,\ -24a^2b$

3. Factoring by Removing the Greatest Common Factor

Since $6a^2b(3a+4b) = 18a^3b + 24a^2b^2$, factoring $18a^3b + 24a^2b^2$ gives $6a^2b(3a+4b)$: the GCF has been removed from every term, leaving a polynomial factor behind.

Negative leading coefficient. When factoring a polynomial whose leading coefficient is negative, remove a *negative* common factor so the bracket starts with a positive term: $-4x^2 + 10x = -2x(2x-5)$. Every sign inside the bracket flips. (This is the convention used in every answer key in this unit.)

Example 2 — Filling in a missing factor

Complete the factoring in each case.

(a) $15x + 25y = _____(3x + 5y)$

(b) $8x^3 + 20x^2 + 4x = _____(2x^2 + 5x + 1)$

(c) $6a^2 + 30a = 6a(a + _____)$

(d) $24p - 16q = 8(_____ - 2q)$

(e) $-5mn - 15m^2 = -5m(_____)$

(f) $21x^2y^2 - 35xy + 7x = 7x(_____)$

Example 3 — Factor completely by removing the GCF

Factor each polynomial completely by removing the greatest common factor.

(a) $18x - 8$

(b) $20x^4 + 5x^2$

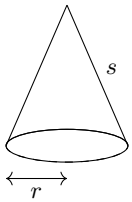
(c) $12a^3b^2 + 9ab^3 + 3ab^4$

(d) $14p^3 - 7p^2 + 21p$

(e) $35xy^2z^3 - 28x^2y^4z^2 + 42x^3y^2z^5$

Example 4 — A formula in factored form

The surface area of a cone is given by $A = \pi r^2 + \pi rs$, where r is the radius of the base and s is the slant height.



(i) Determine the surface area of a cone, to the nearest 0.01 cm^2, with slant height 8.20 cm and base radius 3.10 cm.

(ii) Write the formula for A in factored form.

(iii) Calculate the surface area, to the nearest 0.01 cm^2, using the factored form of A.

Complete Homework Questions 1–18

4. Extension — Monomial Common Factors Involving Fractions

When a polynomial has fractional coefficients, let the common factor itself be a fraction: ask "what do I multiply this fraction by to rebuild each original term?" Chosen well it leaves a remaining polynomial with whole-number coefficients — for example, $\frac{1}{4}x^2 - 2x = \frac{1}{4}x(x - 8)$.

Example 5 *(Extension)* — Fractional common factors

In each case a common factor has already been removed so that the remaining factor is an integral binomial. Fill in the blanks and check your work by expanding the factored form.

(a) $\frac{1}{2}x^2 + 6x = \frac{1}{2}x(\ \ +\ \ )$

(b) $\frac{1}{5}a^2 - 5a = \frac{1}{5}a(\ \ -\ \ )$

(c) $8x + \frac{4}{5} = \frac{4}{5}(\ \ +\ \ )$

(d) $\frac{1}{3}a^2 - \frac{2}{5}b^2 = \frac{1}{15}(\ \ -\ \ )$

Example 6 *(Extension)* — Removing a fractional GCF

Complete the factoring in each case, filling in the blank(s) so that the remaining factor is an integral polynomial. Check mentally by expanding.

(a) $a - \frac{1}{8}a^2 =$ _____$(8 - a)$

(b) $\frac{1}{3}\pi r^2 - 3\pi r =$ _____$(r - 9)$

(c) $6x^2 + 3x + \frac{3}{7} =$ _____$\left(14x^2 + ____\,x + ____\right)$

Complete Homework Questions 19–23 (Extension)

Lesson 2: Common Factors: Binomial Common Factors and Grouping

The common factor can be a whole bracket — and pairing up the terms of a polynomial can create one.

In Lesson 1 the common factor was a monomial: $6x^2 + 9x = 3x(2x + 3)$. Exactly the same idea works when the common factor is a whole *binomial*, and splitting a four-term polynomial into two pairs is a way of *making* a common binomial appear.

1. Factoring Out a Binomial Common Factor

The common factor may be a whole binomial rather than a single term; pull that bracket out exactly as you would a monomial. In $5x(x+2) - 3(x+2)$ the quantity $(x+2)$ multiplies both $5x$ and -3, so $5x(x+2) - 3(x+2) = (x+2)(5x-3)$.

Example 1 — Removing a common binomial factor

Factor each polynomial by removing the common binomial factor.

(a) $(5x)(x+8) - (2)(x+8)$

(b) $5(2-3y) + 4y(2-3y)$

(c) $8a(3a+1) + (3a+1)$

After the binomial is removed, check whether the leftover bracket has a common factor of its own.

Example 2 — Then check the leftover bracket

Factor each polynomial and write your answer in simplest factored form.

(a) $(4y+3)(6y+1) + (4y+3)(2y)$

(b) $5a(a-4) - 15(a-4)$

(c) $3x(x-7) + 7(7-x)$

Complete Homework Question 1

2. Factoring by Grouping

A four-term polynomial usually has no factor common to all four terms, but splitting it into two pairs and removing each pair's own common factor can leave the *same* binomial in both. Pair up, pull a monomial out of each pair, then pull out the shared binomial: this is **factoring by grouping**. For example,

$$x^2 + 3x + 4x + 12 = x(x+3) + 4(x+3) = (x+3)(x+4).$$

Watch for a sign that must be flipped before the binomials match: $(p-q)$ and $(q-p)$ are opposites, so $-k(q-p) = k(p-q)$. **You may need to rearrange the terms before grouping**: if the first pairing does not leave matching binomials, regroup so that terms sharing a factor sit together (Example 3(e)).

Example 3 — Pair up, pull out, pull out again

Factor each polynomial by grouping.

(a) $x^2 + 4x + 5x + 20$ **(b)** $6x^2 - 3x + 8x - 4$ **(c)** $10a^2 - 15a - 4a + 6$

(d) $mn + mp - qn - qp$ **(e)** $3x^2 + 10y^2 - 6xy^2 - 5x$

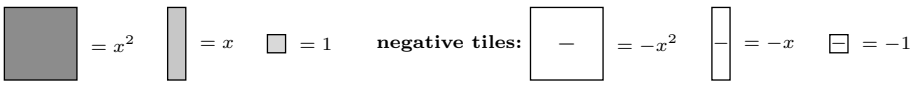
Complete Homework Questions 2–9

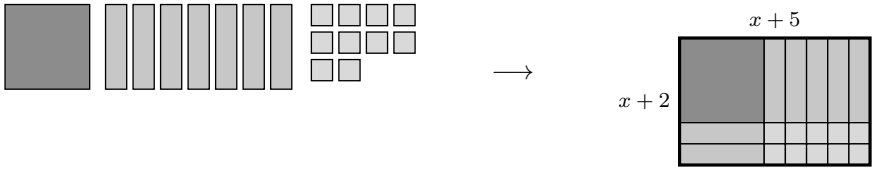
Lesson 3: Factoring Trinomials of the Form $x^2 + bx + c$

Algebra tiles, then inspection: two integers whose product is c and whose sum is b.

1. Why Algebra Tiles Work

With **algebra tiles**, an x^2-tile is a square of side x, an x-tile is a $1 \times x$ rectangle, and a unit tile is a 1×1 square. The collection below — one x^2-tile, seven x-tiles and ten unit tiles — is the polynomial $x^2+7x+10$, and those same tiles rearrange with no gaps or overlaps into a rectangle. Every tile picture in this unit uses the same colour key (negative tiles are used from Section 4 on):

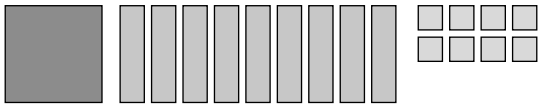




The rectangle's area is the original polynomial, so its length and width are the two binomial factors: length $x + 5$, width $x+2$, giving $x^2+7x+10 = (x+2)(x+5)$. Multiplying back out confirms it: $(x+2)(x+5) = x^2+5x+2x+10 = x^2+7x+10$. ✓

Example 1 — From a pile of tiles to a factored form

Consider the collection of algebra tiles shown below.



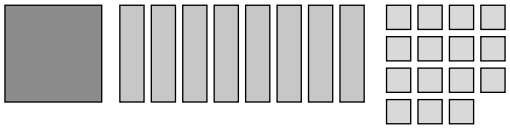
(a) Write the polynomial expression represented by the tiles.

(b) Rearrange the tiles into a rectangle, and state the length and width.

(c) Use your results to write the polynomial in factored form.

Example 2 — The same process, more unit tiles

Consider the collection of algebra tiles shown below.



(a) Write the polynomial expression represented by the tiles.

(b) Rearrange the tiles into a rectangle, and use the result to write the polynomial in factored form.

2. Finding the Pattern: Sum and Product

Example 3 — Expand first, then read the pattern backwards

(a) Expand each pair of binomials.

$(x+3)(x+4) =$ $(x+1)(x+9) =$ $(x+6)(x+6) =$

$(x+7)(x+3) =$ $(x-4)(x-3) =$ $(x+9)(x-5) =$

In each case, how are the coefficient b and the constant term c related to the two numbers being added to x?

(b) Use FOIL to expand $(x+p)(x+q)$ and show that it can always be written in the form $x^2+(p+q)x+pq$.

Part (b) proves in general what part (a) suggested: **to factor x^2+bx+c, find two integers whose sum is b and whose product is c.**

Example 4 — Hunting for the pair

Complete each table by finding two integers with the given sum and the given product. The first row of Table A has been done for you.

Table A		
Sum	**Product**	**Integers**
15	50	5, 10
13	40	
9	18	
6	9	

Table B		
Sum	**Product**	**Integers**
-10	21	
-13	12	
1	-12	
-22	40	

Notice. A **positive** product means the two integers are both positive or both negative; a **negative** product means one of each. Reading that off the signs of b and c:

Sign of c	**Sign of** b	**The two integers**
$c > 0$	$b > 0$	both positive
$c > 0$	$b < 0$	both negative
$c < 0$	$b \neq 0$	one positive and one negative; the one further from 0 carries the sign of b
$c < 0$	$b = 0$	k and $-k$ — this is a **difference of squares** (Lesson 4)

Complete Homework Questions 1–3

3. Factoring $x^2 + bx + c$ by Inspection

To factor $x^2 + bx + c$, find two integers with product c and sum b. If no such pair exists, the trinomial **cannot be factored over the integers**. If the terms share a common factor, remove it *first* — factoring is not complete until no further factoring is possible.

Example 5 — Positive c, including two with a common factor

Factor each expression completely.

(a) $x^2 + 9x + 14$ **(b)** $x^2 + 15x + 14$ **(c)** $x^2 - 15x + 14$

(d) $m^2 - 13m + 12$ **(e)** $y^2 + 4y + 3$ **(f)** $x^2 + 21x + 20$

(g) $3x^2 - 30x + 48$ **(h)** $2x^3 + 24x^2 + 40x$

Negative c works the same way; only now one integer is positive and the other negative. For $x^2 - 2x - 15$ we need product -15 and sum -2 — these are -5 and 3, so $x^2 - 2x - 15 = (x - 5)(x + 3)$.

Example 6 — Negative c

Factor each trinomial by inspection.

(a) $x^2 - x - 20$ **(b)** $x^2 + 5x - 24$ **(c)** $a^2 - 9a - 10$

Example 7 — Every case at once (one of these does not factor)

Factor where possible.

(a) $a^2 + 8a - 33$

(b) $3t^2 - 24t + 45$

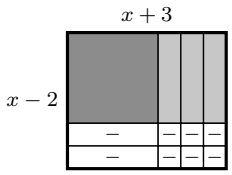
(c) $x^2 - 5x - 9$

(d) $3x^4 - 12x^3 - 36x^2$

Complete Homework Questions 5–14

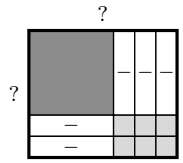
4. Modelling Negative Terms with Algebra Tiles

A **negative** tile is drawn white with a minus sign (see the key in Section 1). Build the rectangle exactly as before: the x^2-tile in the corner, the x-tiles along the top and the side, and the unit tiles filling the corner. The sign of each tile in the filled corner is the product of the signs along its row and column, so *positive × negative gives negative unit tiles.* For $x^2 + x - 6$: three positive x-tiles across the top and two negative x-tiles down the side give $+3x - 2x = +x$, and the corner holds $3 \times 2 = 6$ negative unit tiles. Building it from a pile with only one x-tile needs *zero pairs*: add two positive and two negative x-tiles (worth 0 together) so the rectangle can be completed.



Top: $x + 3$. Side: $x - 2$.
Tiles: x^2, $+3x - 2x = +x$, -6.
So $x^2 + x - 6 = (x + 3)(x - 2)$.
The pair $+3$ and -2: product -6, sum $+1$ — the same rule as before.

Example 8 — Reading and building rectangles with negative tiles



(a) Write the trinomial modelled by the rectangle, and its length and width.

(b) Write the trinomial in factored form, and verify by expanding.

(c) Sketch the tile rectangle for $x^2 + 2x - 8$ and use it to factor the trinomial.

Complete Homework Question 4

5. Trinomials of the Form $x^2 + bxy + cy^2$

A trinomial such as $x^2 + 10xy + 24y^2$ factors exactly like $x^2 + 10x + 24$: the second variable rides along with the second term of each factor. If $x^2 + bx + c = (x + p)(x + q)$, then $x^2 + bxy + cy^2 = (x + py)(x + qy)$, since p and q still only have to multiply to c and add to b.

Example 9 — Expanding shows why the y tags along

(a) Expand $(x+4y)(x+6y)$.

(b) Use your answer to part (a) to factor $x^2+10xy+24y^2$ without any extra work.

Example 10 — Two variables, one with a common factor

Factor each trinomial completely.
(a) $x^2+14xy+45y^2$ **(b)** $x^2+55xy-56y^2$ **(c)** $2a^2-10ab-168b^2$

Complete Homework Questions 15–23

Lesson 4: Difference of Squares

The middle coefficient is 0 — $A^2-B^2=(A-B)(A+B)$, *and why a* sum *of squares never factors.*

1. Where the Pattern Comes From

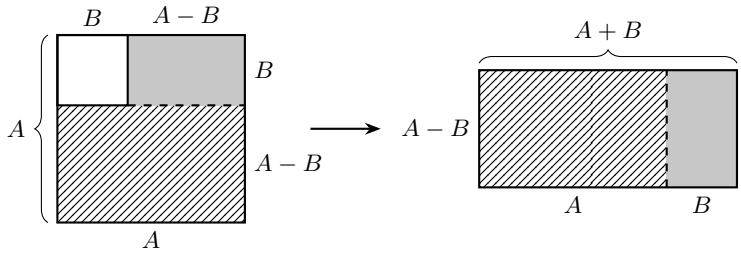
A **difference of squares** such as x^2-49 is the case of x^2+bx+c whose **middle coefficient is** 0, with c the negative of a perfect square. By the sum–product method we need two integers with sum 0 and product -49 — only 7 and -7 — so

$$x^2-49=x^2+(7)x+(-7)x-49=(x+7)(x-7).$$

In general, if $c=-k^2$ the pair is always k and $-k$, which gives the identity

$$\textbf{Difference of squares:}\qquad A^2-B^2=(A-B)(A+B).$$

(Capital letters are used so they don't clash with the b and c of x^2+bx+c.)
Why it works geometrically. Remove a $B\times B$ square from the corner of an $A\times A$ square; the shaded L-shape has area A^2-B^2. Cut along the dashed line into a striped piece (A by $A-B$) and a grey piece ($A-B$ by B). Turn the grey piece and slide it onto the end of the striped piece: the two form a rectangle $A-B$ by $A+B$, so the same area is also $(A-B)(A+B)$.



shaded area $=A^2-B^2$ area $=(A-B)(A+B)$

Example 1 — The bare pattern

Factor each difference of squares.
(a) w^2-9 **(b)** t^2-121 **(c)** m^2-n^2 **(d)** k^2-36

Important. A *sum* of squares such as x^2+16 can **never** be factored over the integers: we would need two integers with positive product and zero sum, but a positive product forces the same sign, so no such pair exists.

2. Substituting Expressions for A and B

In $A^2-B^2=(A-B)(A+B)$, the letters A and B may stand for numbers, variables, monomials or larger expressions. Rewriting $25x^2-64$ as $(5x)^2-(8)^2$ matches the pattern with $A=5x$ and $B=8$, so $25x^2-64=(5x-8)(5x+8)$. Check *both* terms first: if either is not a perfect square, the expression is not a difference of squares.

Example 2 — Coefficients, two variables, and one impostor

Factor, if possible, using the difference of squares pattern.

(a) $25x^2 - 64$ **(b)** $49a^2 - 1$ **(c)** $121 - y^2$

(d) $16p^2 - 9q^2$ **(e)** $9x^2 + 16$

3. Removing a Common Factor First

Always look for a **greatest common factor** first: an expression often becomes a recognizable difference of squares only *after* it is removed, and an answer with a common factor still inside a bracket is not fully factored. Likewise, check each new bracket: it may *itself* be a difference of squares, as in $x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x^2 + 1)$ (Lesson 6).

Example 3 — GCF first, then the pattern

Factor completely by first removing the greatest common factor.

(a) $3a^2 - 75$ **(b)** $5x^2 - 20y^2$

(c) $98p^2q^2 - 2$ **(d)** $2x^3 - 32x$

4. An Application

Example 4 — A rink whose area is a difference of squares

The surface of a rectangular skating rink has area $25m^2 - 4n^2$ square metres.

(a) Write expressions for the length and width of the rink.

(b) The perimeter of the rink is 120 metres. Form an equation in m and n and solve for m.

(c) Determine the length and width of the rink if the length is 50% greater than the width.

Lesson 5: Factoring $ax^2 + bx + c$ (where $a \neq 1$)

Algebra tiles, the area (box) model, and the method of decomposition — and why the pair multiplies to ac.

1. Warm-Up: Multiplying Binomials

Factoring $ax^2 + bx + c$ reverses the multiplication of two binomials, neither of which now has a leading coefficient of 1.

$$(3x + 2)(2x + 5) = 6x^2 + 15x + 4x + 10 = 6x^2 + 19x + 10 \qquad \text{so} \qquad 6x^2 + 19x + 10 \text{ factors to } (3x + 2)(2x + 5)$$

$$(4x - 1)(3x + 2) = 12x^2 + 8x - 3x - 2 = 12x^2 + 5x - 2 \qquad \text{so} \qquad 12x^2 + 5x - 2 \text{ factors to } (4x - 1)(3x + 2)$$

The reverse problem — find two binomials whose product is $6x^2 + 19x + 10$ — is what this lesson solves: first with algebra tiles, then with an area (box) model, then by decomposition.

2. Factoring $ax^2 + bx + c$ Using Algebra Tiles

The tiles use the colour key from Lesson 3. The method is the same as for $x^2 + bx + c$: put the x^2-tiles together in a corner, line x-tiles up along the top and the side, and fill the corner with unit tiles; the rectangle's length and width are the factors.

Example 1 — Reading a trinomial off a pile of tiles

(a) Write a polynomial expression for the group of algebra tiles shown.

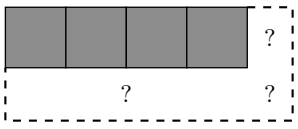


(b) Arrange the algebra tiles into a rectangle and state the length and width of the rectangle.

(c) Use the algebra tile diagram to express the polynomial in factored form.

Example 2 — Building the rectangle yourself

Factor $4x^2 + 13x + 3$ using algebra tiles. Start from the frame below: the four x^2-tiles sit in a row in the corner. Which tiles go along the right edge, and which along the bottom?



3. Reminder: Factoring by Grouping

From Lesson 2: group the first two terms and the last two, remove the common factor from each group, then remove the common binomial — e.g. $10x^2 + 4x + 15x + 6 = 2x(5x + 2) + 3(5x + 2) = (5x + 2)(2x + 3)$. Decomposition (Section 5) is just a way of *creating* the four terms that grouping needs.

4. The Area (Box) Model

Tiles become tedious when a, b and c are large, and they need negative tiles when b or c is negative. An **area box** works in every case. Draw a 2×2 grid, put ax^2 in the top-left cell and c in the bottom-right cell. The other two cells hold the middle term bx split into two pieces whose **product is** $ac \cdot x^2$ and whose **sum is** bx. Then take the GCF of each row and each column: those are the factors. For $6x^2 - x - 2$, $ac = -12$ and the pair is -4 and 3:

	$3x$	-2
$2x$	$6x^2$	$-4x$
$+1$	$3x$	-2

Row 1: $6x^2$ and $-4x$ share $2x$. Row 2: $3x$ and -2 share 1.
Columns: $3x$ and -2.
Choose each row's sign so that the columns agree.
So $6x^2 - x - 2 = (2x + 1)(3x - 2)$. Check: $6x^2 - 4x + 3x - 2 = 6x^2 - x - 2$ ✓

Each row of the box is one group in factoring by grouping: $6x^2 - 4x + 3x - 2 = 2x(3x - 2) + 1(3x - 2)$.

Example 3 — Fill in the box

Complete each area box and write the trinomial in factored form. Verify by expanding.

(a) $3x^2 + 10x - 8 \quad (ac = -24)$

$3x^2$	
	-8

(b) $2x^2 - 7x + 6 \quad (ac = 12)$

$2x^2$	
	6

Complete Homework Questions 1–3

5. The Method of Decomposition — and Why the Pair Multiplies to ac

Why ac? Expand two general binomials *without* collecting the middle terms:

$$(px + q)(rx + s) = \underbrace{pr}_{a}\, x^2 + \underbrace{ps\,x + qr\,x}_{\text{the two middle pieces}} + \underbrace{qs}_{c}.$$

The two middle coefficients add to $b = ps + qr$, and they *multiply* to $(ps)(qr) = (pr)(qs) = ac$. So the middle term of every factorable $ax^2 + bx + c$ splits into two pieces whose product is ac and whose sum is b. (When $a = 1$ this is exactly the Lesson 3 rule: the product is $1 \cdot c = c$.)

The method of factoring $ax^2 + bx + c$ by splitting the value of b into two integers whose **product is** ac and whose **sum is** b is called the **method of decomposition**.

Every decomposition has the same five steps: **(1)** compute ac; **(2)** find the pair of integers with product ac and sum b; **(3)** split the middle term with that pair; **(4)** group and remove the common binomial factor. **(5)** Verify by expanding.

Example 4 — Decomposition, checked against the tiles

Factor the following using the method of decomposition, verify by expanding, and compare your answers with Example 1 and Example 2.

(a) $3x^2 + 8x + 4$

(b) $4x^2 + 13x + 3$

6. Finding the Pair When ac Is Large

List factor pairs systematically. Start at $1 \times |ac|$ and work up, stopping when the pair reaches $\sqrt{|ac|}$. If $ac < 0$ the two numbers have opposite signs and you want a *difference* of $|b|$; if $ac > 0$ they have the same sign as b.

Calculator help (TI-30XIIS). Divide to find each partner: $|ac| \div$ a trial number gives the other number of the pair (e.g. $180 \div 12 = 15$). Stop at $\sqrt{|ac|}$ — press 2nd x^2 for the square root.

Calculator check. Substitute the same value of x (try 2 or 3, not 0 or 1) into the trinomial and into your factored form; the two results must be equal. Two different quadratics can agree for at most two values of x, so checking two values is a strong test.

Example 5 — A large ac

Factor $18x^2 + 3x - 10$. Use a factor-pair list to find the pair, then verify on the calculator by substituting a value of x.

7. Practice: Putting It All Together

Decomposition also handles a negative leading term (rearrange into descending order, then remove a negative common factor so the bracket starts with a positive term, as in Lesson 1), perfect-square trinomials, and two-variable trinomials $ax^2 + bxy + cy^2$.

Always look for a common factor first. Removing it before decomposing leaves a much smaller ac: $10a^2 + 25a - 15 = 5(2a^2 + 5a - 3)$, and $6x^3 + 5x^2 + x = x(6x^2 + 5x + 1)$ turns a cubic into a trinomial. A factorization is not finished until no bracket factors any further.

Example 6 — The full range of cases

Factor.

(a) $6x^2 + 19x - 7$ **(b)** $4n^2 - 5n - 6$ **(c)** $9x^2 - 12x + 4$

(d) $20 - 3y - 2y^2$ **(e)** $3x^2 - 7xy + 2y^2$

Complete Homework Questions 4–16

Lesson 6: Further Factoring

Perfect square trinomials, trinomials in a monomial disguise, and extended differences of squares. (Extension: higher-degree disguises.)

1. Perfect Square Trinomials

Squaring a binomial gives

$$(p+q)^2 = p^2 + 2pq + q^2 \qquad (p-q)^2 = p^2 - 2pq + q^2$$

A trinomial of this form is a **perfect square trinomial**. Read backwards, the pattern gives a quick test.

A trinomial is a **perfect square trinomial** exactly when **(1)** the *first* term is a perfect square, **(2)** the *last* term is a perfect square (and positive), and **(3)** the *middle* term is **twice the product** of the square roots of the first and last terms (with the matching sign).

All three conditions must hold. In $m^2 + 14m + 49$ the first term m^2 and the last term $49 = 7^2$ are perfect squares, and the middle term $14m = 2(m)(7)$ is exactly twice their product — so $m^2 + 14m + 49 = (m + 7)^2$.

Example 1 — Applying the three-part test

Which of the following are perfect square trinomials?

(a) $b^2 + 6b + 9$ **(b)** $x^2 - 11x + 8$

(c) $9x^2 - 42x + 49$ **(d)** $y^2 + 6y - 9$

Example 2 — Reading the pattern backwards to fill a gap

Fill in each blank so that the trinomial is a perfect square trinomial.

(a) $x^2 + _____ + 121$ **(b)** $x^2 - _____ + 169$ **(c)** $16x^2 + _____ + 81$ **(d)** $25m^2 + 40m + _____$

Example 3 — Factoring a perfect square trinomial

Factor each perfect square trinomial.

(a) $81x^2 - 18x + 1$ **(b)** $9 + 24x + 16x^2$ **(c)** $\frac{1}{4}a^2 - 3ab + 9b^2$

2. Trinomials of the Form $u^2 + bu + c$ where u Is a Monomial

In $x^2 + bx + c$ the terms have degrees $2, 1, 0$. Inspection still works when the degrees are $4, 2, 0$: the variable x has simply been replaced by the monomial x^2 (or by a product such as pq). (Degrees $6, 3, 0$ and higher work the same way — see the Extension in Section 5.)

Key idea: treat the squared monomial as a single block. To factor m^4+7m^2+12, notice it can be written $(m^2)^2+7(m^2)+12$. Substituting $u = m^2$ turns this into the familiar trinomial $u^2 + 7u + 12 = (u + 3)(u + 4)$. Replacing u by m^2 again gives $m^4 + 7m^2 + 12 = (m^2 + 3)(m^2 + 4)$. With practice the substitution can be done mentally, treating m^2 exactly as x was treated in earlier lessons — but write the substitution down until it is automatic, or the final brackets end up holding the wrong variable.

Example 4 — Substitute, factor, substitute back

Factor completely.
(a) $b^4 - 3b^2 - 40$ **(b)** $y^4 + 3y^2 - 4$

3. Trinomials of the Form $au^2 + bu + c$ where u Is a Monomial

The same disguise works with decomposition when the leading coefficient is not 1. There are two equivalent ways to handle $6m^4 - 7m^2 - 3$.

Method 1: Substitution

Let $u = m^2$:

$$6m^4 - 7m^2 - 3 = 6u^2 - 7u - 3$$

Decompose: $6u^2 - 9u + 2u - 3$

$$= 3u(2u - 3) + 1(2u - 3) = (2u - 3)(3u + 1)$$

Replace u by m^2:

$$= (2m^2 - 3)(3m^2 + 1)$$

Method 2: Direct decomposition

Split the middle term directly in terms of m^2:

$$6m^4 - 7m^2 - 3 = 6m^4 - 9m^2 + 2m^2 - 3$$

$$= 3m^2(2m^2 - 3) + 1(2m^2 - 3)$$

$$= (2m^2 - 3)(3m^2 + 1)$$

Example 5 — Decomposition in disguise

Factor completely.
(a) $10x^4 - 3x^2 - 4$ **(b)** $3p^2q^2 - 20pq + 12$

(c) $12x^4 + 8x^2 - 15$

4. More Complex Differences of Squares

The pattern $A^2 - B^2 = (A - B)(A + B)$ also holds when A and B are themselves polynomials.

To factor $m^4 - 81n^4$, rewrite it as a difference of two squares: $m^4 - 81n^4 = (m^2)^2 - (9n^2)^2 = (m^2 - 9n^2)(m^2 + 9n^2)$. The first factor, $m^2 - 9n^2$, is *itself* a difference of squares, so it factors again: $= (m - 3n)(m + 3n)(m^2 + 9n^2)$. Always check whether a newly formed factor can be split further — "factor completely" means no stopping until every common factor has been removed and every possible difference of squares has been used.

Example 6 — Keep going until nothing splits

Factor completely.
(a) $w^4 - 16$ **(b)** $162 - 2x^4$ **(c)** $3p^5q^4 - 48pt^4$

5. Extension — Higher-Degree Disguises

The block can be any power: in $x^6 - 7x^3 - 18$ the degrees are $6, 3, 0$, so the block is x^3; in $z^{10} - 3z^5 - 10$ the block is z^5.

Example 7 ***(Extension)*** **— Bigger blocks**

Factor completely.

(a) $x^6 - 7x^3 - 18$

(b) $z^{10} - 3z^5 - 10$

Lesson 7: Choosing a Factoring Strategy

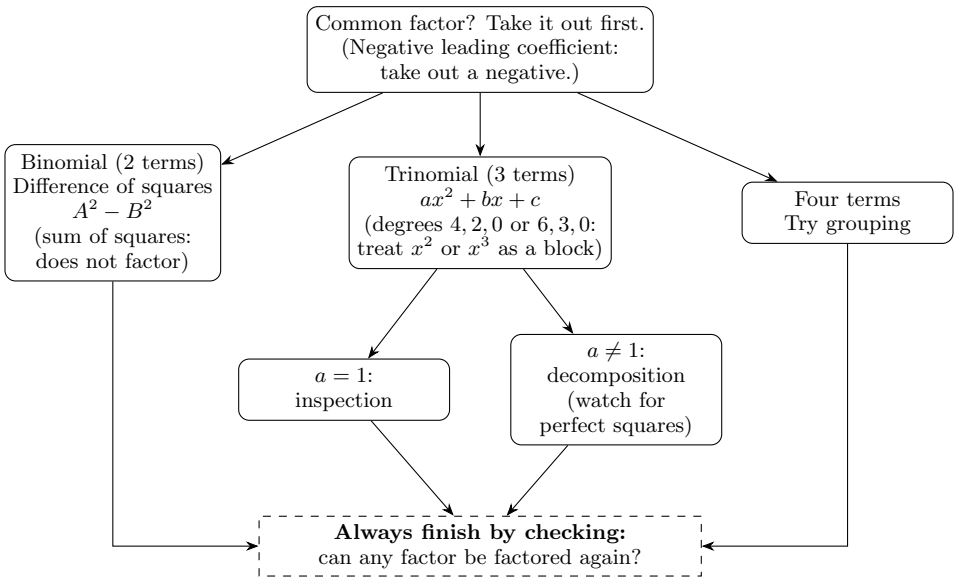
No new technique — a decision process for picking the right tool, then practice on a mixture of expressions.

1. A Strategy for Factoring Any Expression

Work through this checklist, *in order*, every time you are asked to factor.

1. **Common factor first.** Always check whether every term shares a common factor. If so, factor it out before doing anything else — this makes every later step easier. If the **leading coefficient is negative**, remove a *negative* common factor so the bracket starts with a positive term (Lesson 1).
2. **Two terms left (binomial)?** Check whether it is a **difference of squares**, $A^2 - B^2 = (A - B)(A + B)$. A **sum of squares** $A^2 + B^2$ does **not** factor.
3. **Three terms left (trinomial) of the form $x^2 + bx + c$?** Factor **by inspection** (find two integers that multiply to c and add to b). A trinomial in x^2 or x^3 (degrees $4, 2, 0$ or $6, 3, 0$) works the same way: treat x^2 (or x^3) as one block (Lesson 6).
4. **Three terms left (trinomial) of the form $ax^2 + bx + c$ with $a \neq 1$?** Factor by the **method of decomposition** (or an area box). Keep an eye out for **perfect square trinomials** — they decompose very quickly once you spot them. Degrees $4, 2, 0$ or $6, 3, 0$? Treat x^2 or x^3 as one block here too.
5. **Four terms left?** Try **factoring by grouping**.
6. **Always finish with a check.** Look at each factor you ended up with and ask: can *this* be factored again? Many mistakes on tests come from stopping one step too early.

This checklist is often drawn as a flowchart.



2. Working Through the Checklist

In each Example below, decide which line of the checklist applies — possibly more than once — before you factor.

Example 1 — Common factor, then a difference of squares

Factor completely.
(a) $8x^2 - 50$ **(b)** $-12x^2 + 27$

Example 2 — Three terms: inspection or decomposition?

Factor completely.
(a) $x^2 - 2x - 63$ **(b)** $-6x^2 + 5x + 21$

Example 3 — Four terms, and a trinomial in disguise

Factor completely.
(a) $x^2 - xy + 3x - 3y$ **(b)** $x^4 - 13x^2 + 36$

Example 4 — A perfect square, and a decomposition in two variables

Factor completely.
(a) $4a^2 - 20ab + 25b^2$ **(b)** *(Extension)* $6t^6 + 5t^3p^2 - 6p^4$

Lesson 8: Factoring Polynomial Expressions Unit Review

Laid out like the unit test — Beginning, Progressing and Mastery questions for outcome AN5.

1. The Factoring Toolbox

AN5: *Demonstrate an understanding of common factors and trinomial factoring, concretely, pictorially and symbolically.* Every tool from Lessons 1–6 in one place (Lesson 7's checklist tells you which to try first):

Tool	Use it when...	Example
Greatest common factor (1)	every term shares a factor; a negative leading coefficient means take out a negative	$6x^3 - 9x^2 = 3x^2(2x - 3)$
Grouping (2)	four terms	$x^2 + 3x + 4x + 12 = (x + 3)(x + 4)$
Inspection (3)	$x^2 + bx + c$: product c, sum b	$x^2 - x - 12 = (x - 4)(x + 3)$
Decomposition or area box (5)	$ax^2 + bx + c$: product ac, sum b	$6x^2 - x - 2 = (2x + 1)(3x - 2)$
Difference of squares (4)	$A^2 - B^2$ (a sum of squares does not factor)	$9x^2 - 25 = (3x - 5)(3x + 5)$
Perfect square trinomial (6)	$A^2 \pm 2AB + B^2$	$4x^2 - 12x + 9 = (2x - 3)^2$
Trinomial in x^2 (6)	degrees $4, 2, 0$	$x^4 - 5x^2 + 4 = (x - 1)(x + 1)(x - 2)(x + 2)$

Always: factor completely (check every bracket again) and verify by expanding.

2. How This Review Works

The questions are grouped the way the unit test is: **Beginning (BEG)** questions check the basic skills, **Progressing (PRG)** questions use them on trinomials with $a \neq 1$, models and "factor completely" problems, and **Mastery (MAS)** questions ask you to reason, explain errors, work backwards from a factor, or combine several steps. Try every level.

Complete Homework Questions 1–21